CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

FRANK BORGES HEIFER RANCH #2

28533 TWO RIVERS RD

MANTECA CA 95337

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Borges Heifer Ranch #2

Physical address of dairy:

Number and Street	City	County	Zip Code
28533 Two Rivers Rd.	Manteca	San Joaquin	95337

Street and nearest cross street (if no address): Avenue D

Date facility was originally placed in operation: 08/01/1985

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-X050-X030-XXXX

B. OPERATORS

Borges, Frank

Operator name: Borges, Frank

Telephone no.: Landline: ; Cellular: (209) 985-9933

Mailing Address Number and Street	City	State	Zip Code
1518 Thunderbird RD	Modesto	CA	95356

This operator is responsible for paying permit fees.

Silveira, Manuel Sr

Operator name: Silveira, Manuel Sr

Telephone no.: Landline: ; Cellular: (209) 403-3001

Mailing Address Number and Street	City	State	Zip Code
28533 Two Rivers RD	Manteca	CA	95337

C. OWNERS

Silveira, Manuel Sr

Legal owner name: Silveira, Manuel Sr

Telephone no.: Landline: ; Cellular: (209) 403-3001

Mailing Address Number and Street	City	State	Zip Code
28533 Two Rivers RD	Manteca	CA	95337

Annual Report - General Order No. R5-2007-0035
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	590	0	0	0
Maximum number	0	0	590	0	0	0
Average number	0	0	590	0	0	0
Avg live weight (lbs)	0	0	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 6,167.60 *tons per reporting period*

Total nitrogen from manure: 55,991.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 39,193.70 *lbs per reporting period*

Total phosphorus from manure: 9,475.40 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	Corral solids	975.00 *ton*	Dry-weight	34.3		23,300.00	6,200.00	25,500.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	29,850.80	7,943.13	32,669.33	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	29,850.80	7,943.13	32,669.33	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Fields 1,2,3	17	17	2	manure	X257-X050-X017-XXXX X257-X050-X022-XXXX
Totals for areas that were used for application	17	17	2		
Totals for areas that were not used for application					
Land application area totals	17	17	2		

B. CROPS AND HARVESTS

Fields 1,2,3

Field name: Fields 1,2,3

10/18/2022: Oats, hay

Crop: Oats, hay Acres planted: 17 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	56.10 *ton*	Dry-weight		11.9	19,600.00	4,700.00	37,400.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	3.30	113.97	27.33	217.47	715.20

05/22/2023: Sudangrass, hay

Crop: Sudangrass, hay Acres planted: 17 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	125.80 *ton*	Dry-weight		7.8	18,000.00	3,400.00	36,400.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	7.40	245.62	46.40	496.70	1,405.50

NUTRIENT BUDGET

A. LAND APPLICATIONS

Fields 1,2,3 - 10/18/2022: Oats, hay

Field name: Fields 1,2,3

Crop: Oats, hay — Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	145.43	38.70	159.16	0.00	80.75 *ton*
Application event totals		145.43	38.70	159.16	0.00	

10/30/2022	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

Fields 1,2,3 - 05/22/2023: Sudangrass, hay

Field name: Fields 1,2,3

Crop: Sudangrass, hay — Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	321.47	85.54	351.82	0.00	178.50 *ton*
Application event totals		321.47	85.54	351.82	0.00	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

Fields 1,2,3 - 05/22/2023: Sudangrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

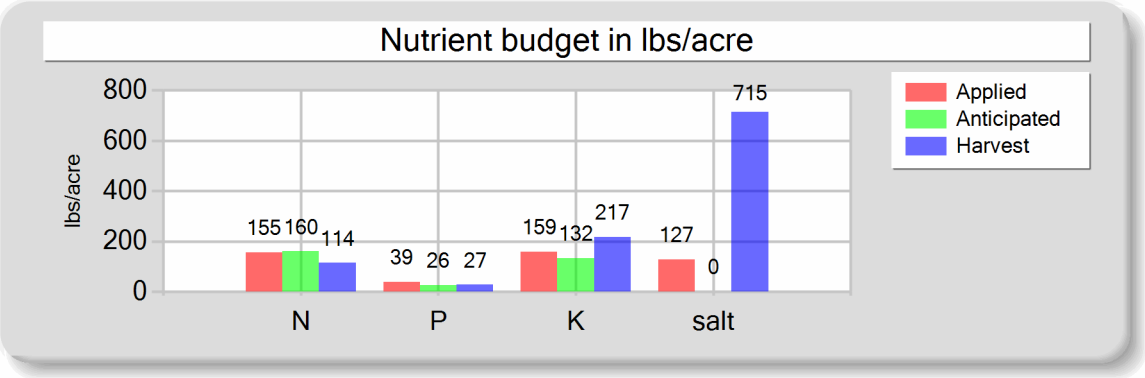
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.58	0.00	0.00	127.46	3,508,857.00 *gal*
Application event totals		2.58	0.00	0.00	127.46	

B. NUTRIENT BUDGET

Fields 1,2,3 - 10/18/2022: Oats, hay

Field name: Fields 1,2,3 Crop: Oats, hay Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	145.43	38.70	159.16	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.58	0.00	0.00	127.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	155.01	38.70	159.16	127.46
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	113.97	27.33	217.47	715.20
Nutrient balance	41.04	11.37	-58.31	-587.74
Applied to removed ratio	1.36	1.42	0.73	0.18

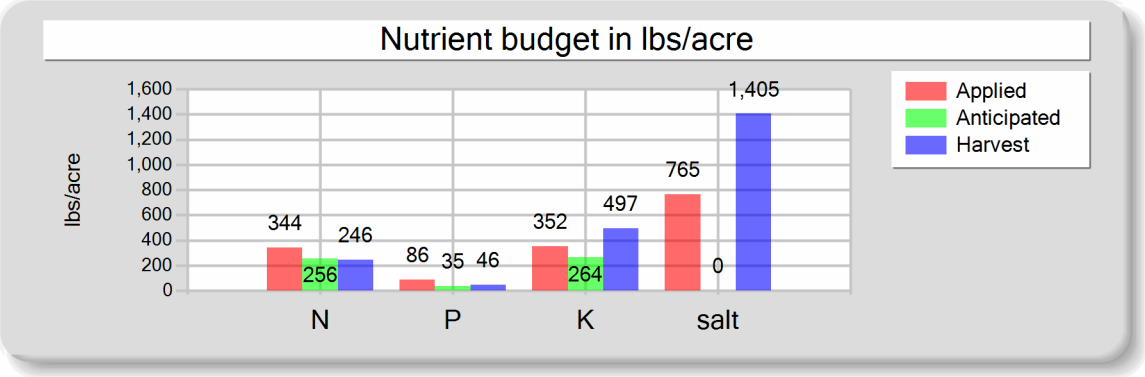
Fresh water applied	
3,508,857.00	*gallons*
129.22	*acre-inches*
7.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fields 1,2,3 - 05/22/2023: Sudangrass, hay

Field name: Fields 1,2,3 Crop: Sudangrass, hay Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	321.47	85.54	351.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.50	0.00	0.00	764.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.97	85.54	351.82	764.76
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	245.62	46.40	496.70	1,405.50
Nutrient balance	98.35	39.15	-144.88	-640.74
Applied to removed ratio	1.40	1.84	0.71	0.54

Fresh water applied	
21,053,142.00	*gallons*
775.32	*acre-inches*
45.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Exported Manure

Sample and source description: Exported Manure

Sample date: 09/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	6,200.00	25,500.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 08/04/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.50	0.00	1.50								108.00	74
DL	0.01	0.01	0.01								10.00	1

Domestic Well

Domestic Well

Domestic Well House

Sample description: Domestic Well House

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								1,110.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Fields 1,2,3 - 10/18/2022: Oats, hay

Field 1, 2, 3 Oat Hay

Sample and source description: Field 1, 2, 3 Oat Hay

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,700.00	37,400.00		12.30
DL	0.01	0.01	0.01		1.00

Fields 1,2,3 - 05/22/2023: Sudangrass, hay

Fields 1,2,3 - 05/22/2023: Sudangrass, hay

Field 1, 2, 3 Sudan Hay

Sample and source description: Field 1, 2, 3 Sudan Hay

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,000.00	3,400.00	36,400.00		10.30
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

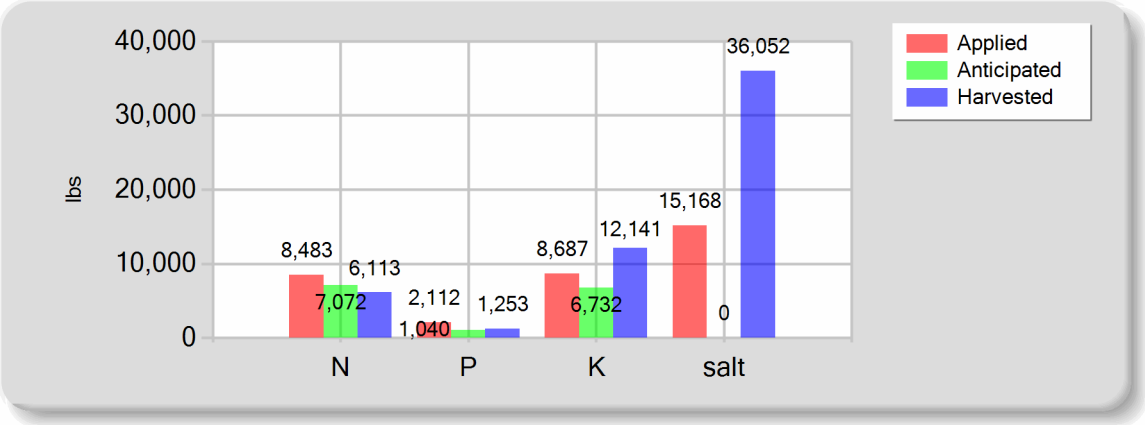
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

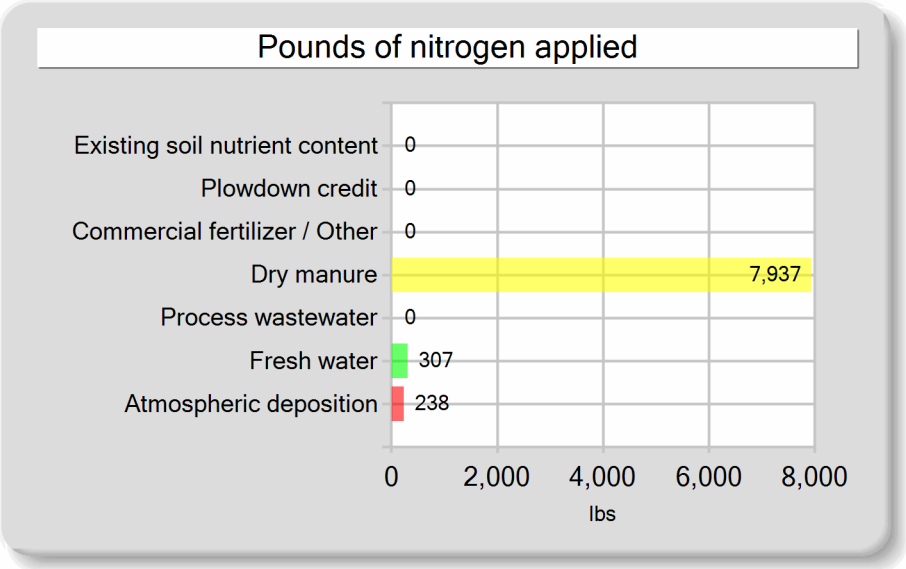
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

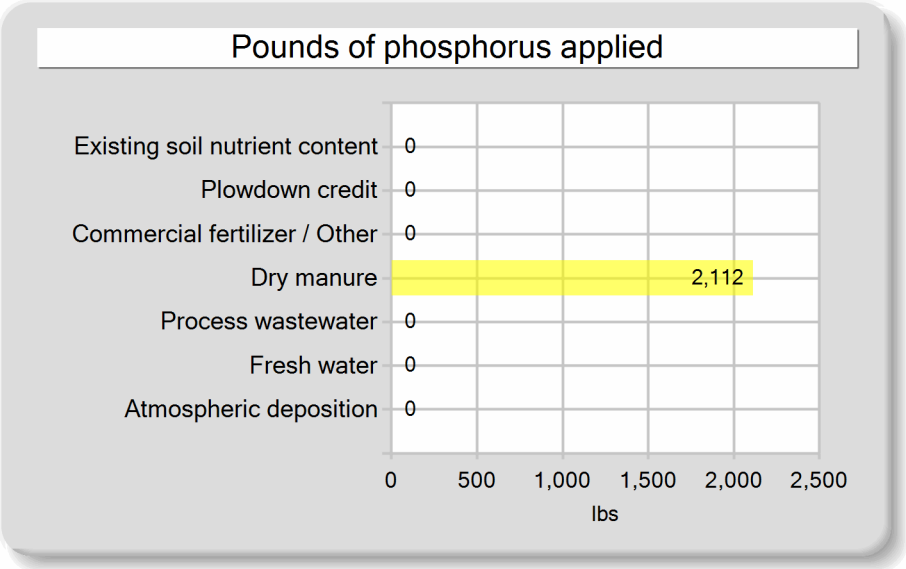
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	7,937.25	2,112.06	8,686.69	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	307.45	0.00	0.00	15,167.77
Atmospheric deposition	238.00	0.00	0.00	0.00
Total nutrients applied	8,482.70	2,112.06	8,686.69	15,167.77
Anticipated crop nutrient removal	7,072.00	1,040.40	6,732.00	0.00
Actual crop nutrient removal	6,112.98	1,253.30	12,140.82	36,051.77
Nutrient balance	2,369.73	858.76	-3,454.13	-20,884.00
Applied to removed ratio	1.39	1.69	0.72	0.42

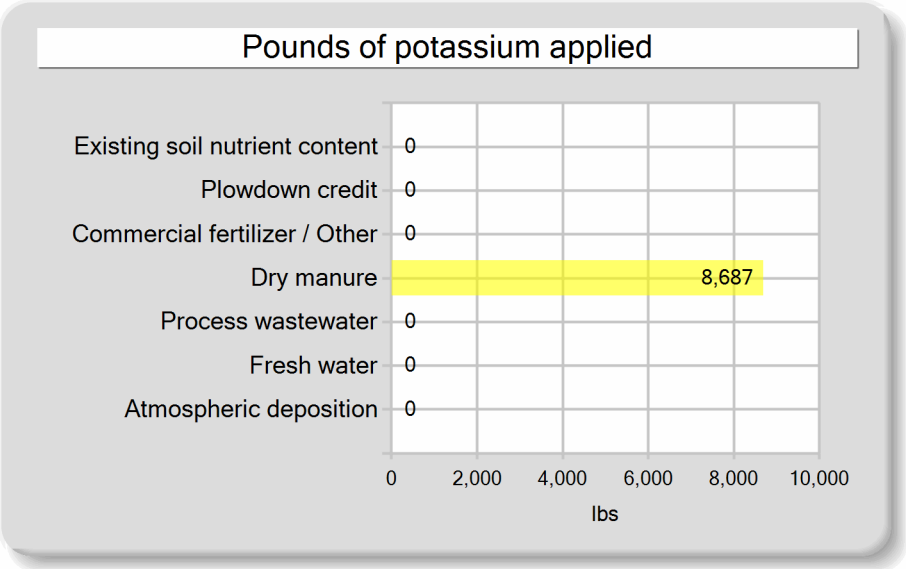
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

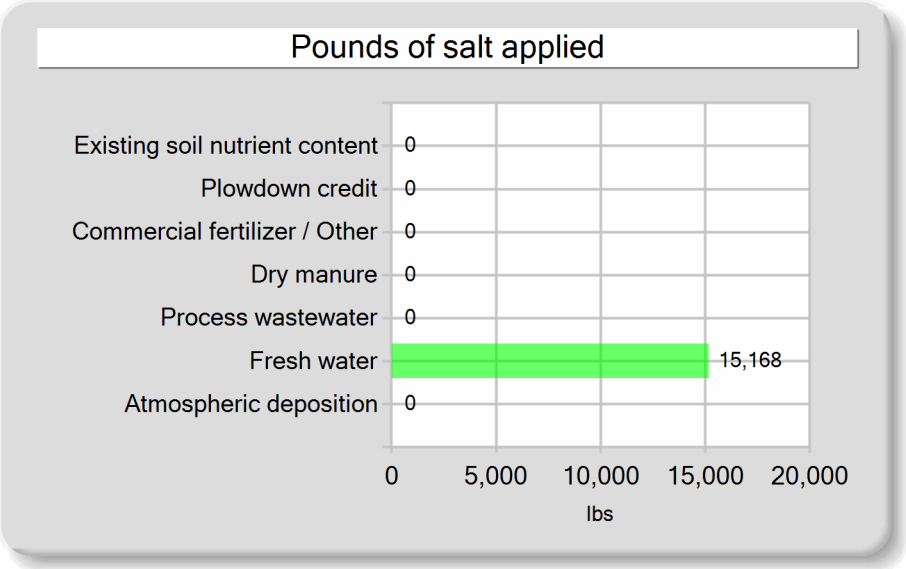


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Facility doesn't house or milk cows so wastewater is minimal and not enough is produced for sampling therefore no samples were taken

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel Silveira, Sr	Frank Borges
PRINT OR TYPE NAME	PRINT OR TYPE NAME
	Jul 3, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Frank Borges

Name of Dairy Facility: Frank Borges Heifer Ranch #2

Facility Address:

28533 Two Rivers Rd.	Manteca	San Joaquin	95337
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Frank Borges (209) 985-9933
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: B and M Hauling

Address of Hauling Company/Person:

11708 E Moncure AVE	Ripon	CA	95366
Number and Street	City	State	Zip Code

Contact Person: Ed Machado (209) 599-3887
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

B and M Farms (209) 599-3887
Name Phone Number

11708 E Moncure AVE	Ripon	CA	95366
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

	Ripon	95366
Address	City	Zip Code

Lorenzen Rd	San Joaquin
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 975.00 tons

Manure Solids Content: 65.7 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Frank Borges 9/18/23

Operator Signature Date

Frank Borges 9/18/23

Hauler Signature Date

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Frank Borges Dairy
1518 Thunderbird Drive
Modesto CA 95356
00-0021379 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0438
Sampled By: M Pedroso
Requested By: Frank Borges
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project: Heifer Ranch #2 - Two Rivers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW House	12/06/23	13:25	1110	1.11	0.1	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Frank Borges Dairy
1518 Thunderbird Drive
Modesto CA 95356
00-0021379 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0438
Sampled By: M Pedroso
Requested By: Frank Borges
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project: Heifer Ranch #2 - Two Rivers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW House	12/06/23	13:25	**1110**	**1.11**	0.1	0.27	0.02

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Frank Borges Dairy
1518 Thunderbird Drive
Modesto, CA 95356

Account# 00-0021379
Account Manager: Ben Nydam
Submitted By: Frank Borges
Ranch: Heifer Ranch #2 - Two Rivers

Received: 12/07/2023 15:35
Reported: 12/13/2023 10:57

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0438-01	DW House	Supply Water	M Pedroso		12/06/2023 13:25

Default Cooler Temperature on Receipt °C: 2.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Received: 12/07/2023 15:35
Reported: 12/13/2023 10:57

Sample Results

Sample: DW House
23L0438-01 (Water)

Sampled: 12/6/2023 13:25
Sampled By: M Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.11**	mmhos/cm	0.01	1		12/11/23 15:30	SM 2510 B		BEL0300
Electrical Conductivity umhos	**1110**	umhos/cm	10.0	1		12/11/23 15:30	SM 2510 B		BEL0300
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 13:25	Field		BEL0453
Nitrate Nitrogen as NO3N	**0.1**	mg/L	0.1	1	10	12/08/23 10:06	EPA 300.0		BEL0243
Temperature	**25.0**	units	0.0	1		12/11/23 15:30	SM 4500-H+	H	BEL0300
pH	**7.6**	units	1.0	1		12/11/23 15:30	SM 4500-H+	H	BEL0300

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Ranch: Heifer Ranch #2 - Two Rivers

Received: 12/07/2023 15:35
Reported: 12/13/2023 10:57

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0243										
Blank (BEL0243-BLK1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0243-BLK2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0243-BLK3)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0243-BLK4)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0243-BS1)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		94.4	90-110		
LCS (BEL0243-BS2)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		95.4	90-110		
LCS (BEL0243-BS3)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.6		0.1	mg/L	5.000		111	90-110		
Duplicate (BEL0243-DUP1)		**Source: 23L0405-01**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L		ND				10
Duplicate (BEL0243-DUP2)		**Source: 23L0412-04**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L		ND				10
Duplicate (BEL0243-DUP3)		**Source: 23L0413-01**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			2.74	10
Matrix Spike (BEL0243-MS1)		**Source: 23L0405-01**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	ND	94.1	90-110		
Matrix Spike (BEL0243-MS2)		**Source: 23L0412-04**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000	ND	95.2	90-110		
Matrix Spike (BEL0243-MS3)		**Source: 23L0413-01**			Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000	0.04	96.2	90-110		
Reference (BEL0243-SRM1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		94.8	90-110		
Reference (BEL0243-SRM2)					Prepared: 12/7/2023 Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		95.2	90-110		

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DELLAVALLE™
LABORATORY INC

Frank Borges Dairy
1518 Thunderbird Drive
Modesto, CA 95356

Account# 00-0021379
Account Manager: Ben Nydam
Submitted By: Frank Borges
Ranch: Heifer Ranch #2 - Two Rivers

Received: 12/07/2023 15:35
Reported: 12/13/2023 10:57

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0243 (Continued)										
Reference (BEL0243-SRM3)					Prepared: 12/7/2023	Analyzed: 12/8/2023				
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		95.2	90-110		
Reference (BEL0243-SRM4)					Prepared: 12/7/2023	Analyzed: 12/8/2023				
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		95.3	90-110		

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Received: 12/07/2023 15:35
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0300										
Blank (BEL0300-BLK1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.4		1.0	units						
Blank (BEL0300-BLK2)					Prepared & Analyzed: 12/11/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.1		1.0	units						
Blank (BEL0300-BLK3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEL0300-DUP1)		**Source: 23L0425-01**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	3.39		0.01	mmhos/cm		3.37			0.650	10
pH	5.8		1.0	units		5.8			0.514	10
Electrical Conductivity umhos	3390		10.0	umhos/cm		3370			0.650	10
Duplicate (BEL0300-DUP2)		**Source: 23L0442-01**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.49		0.01	mmhos/cm		0.48			1.55	10
Electrical Conductivity umhos	489		10.0	umhos/cm		481			1.55	10
pH	7.7		1.0	units		7.7			0.260	10
Reference (BEL0300-SRM1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	433			umhos/cm	426.0		102	90-110		
Reference (BEL0300-SRM2)					Prepared & Analyzed: 12/11/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0300-SRM3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0300-SRM4)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0300-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		

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LABORATORY INC

Frank Borges Dairy
1518 Thunderbird Drive
Modesto, CA 95356

Account# 00-0021379
Account Manager: Ben Nydam
Submitted By: Frank Borges
Ranch: Heifer Ranch #2 - Two Rivers

Received: 12/07/2023 15:35
Reported: 12/13/2023 10:57

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0300 (Continued)										
Reference (BEL0300-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0300-SRM6)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		99.8	97.5-102.5		
Reference (BEL0300-SRM7)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		99.5	97.5-102.5		
Reference (BEL0300-SRM8)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/07/23 15:35 23L0438

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21379 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Borges, Frank Dairy
Address: 1518 Thunderbird Drive
City, State, Zip: Modesto,CA 95356
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Frank Borges - (209) 985-9933

Ranch: Heifer Ranch #2 - Two Rivers

Date sampled: 12/6/23

Sampled by: mpedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-House	Sampled From:	12/6/23	1:25p	[redacted] Ø	21
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	mpedroso	CASInc	12/6/23 125p	12/7/23 11A
Second	JLF	DLV	12-7-23 11:00	12-7-23 5:15
Third				
Fourth	[signature]	DLI	12-7-23 15:35	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☐ Blue Ice ☑ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/07/23 15:35 23L0438

DLI SEND OUT